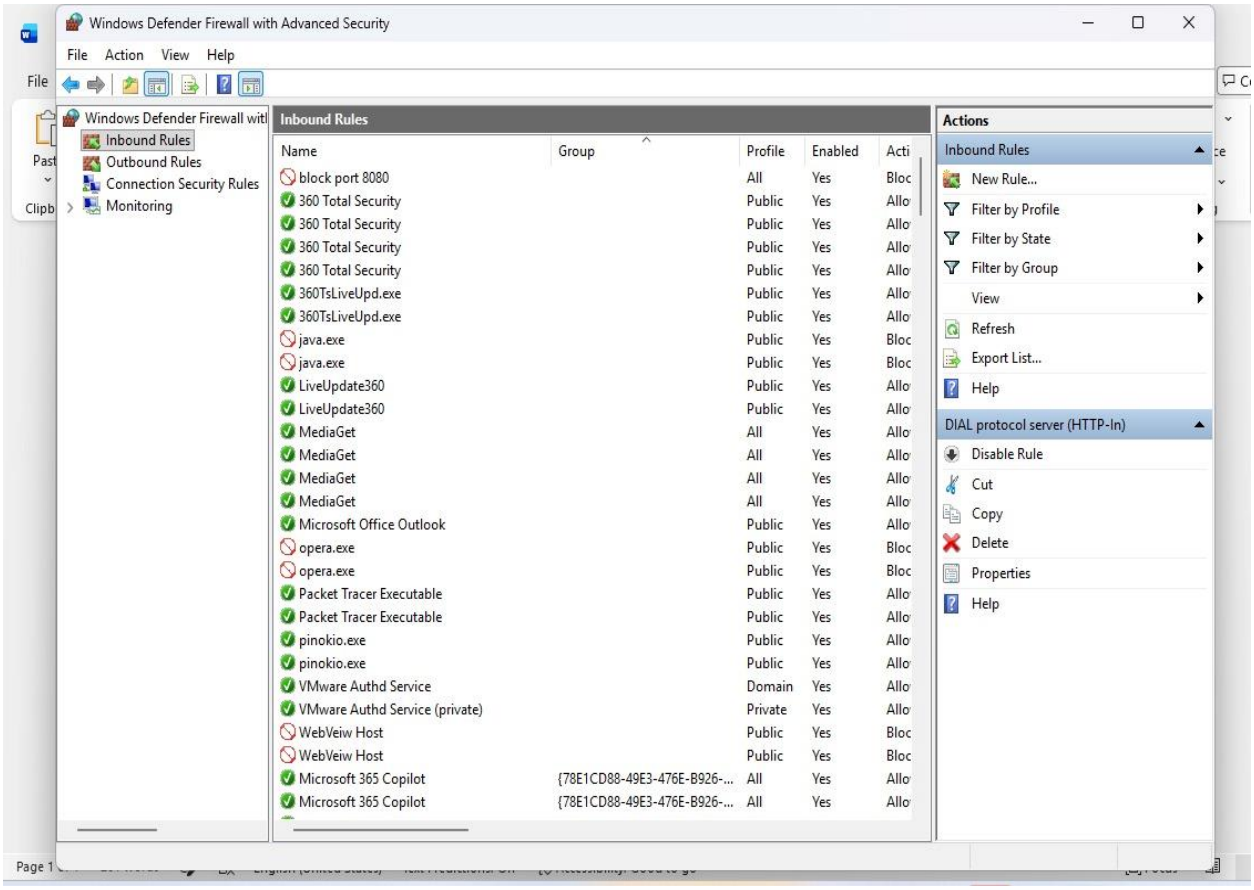
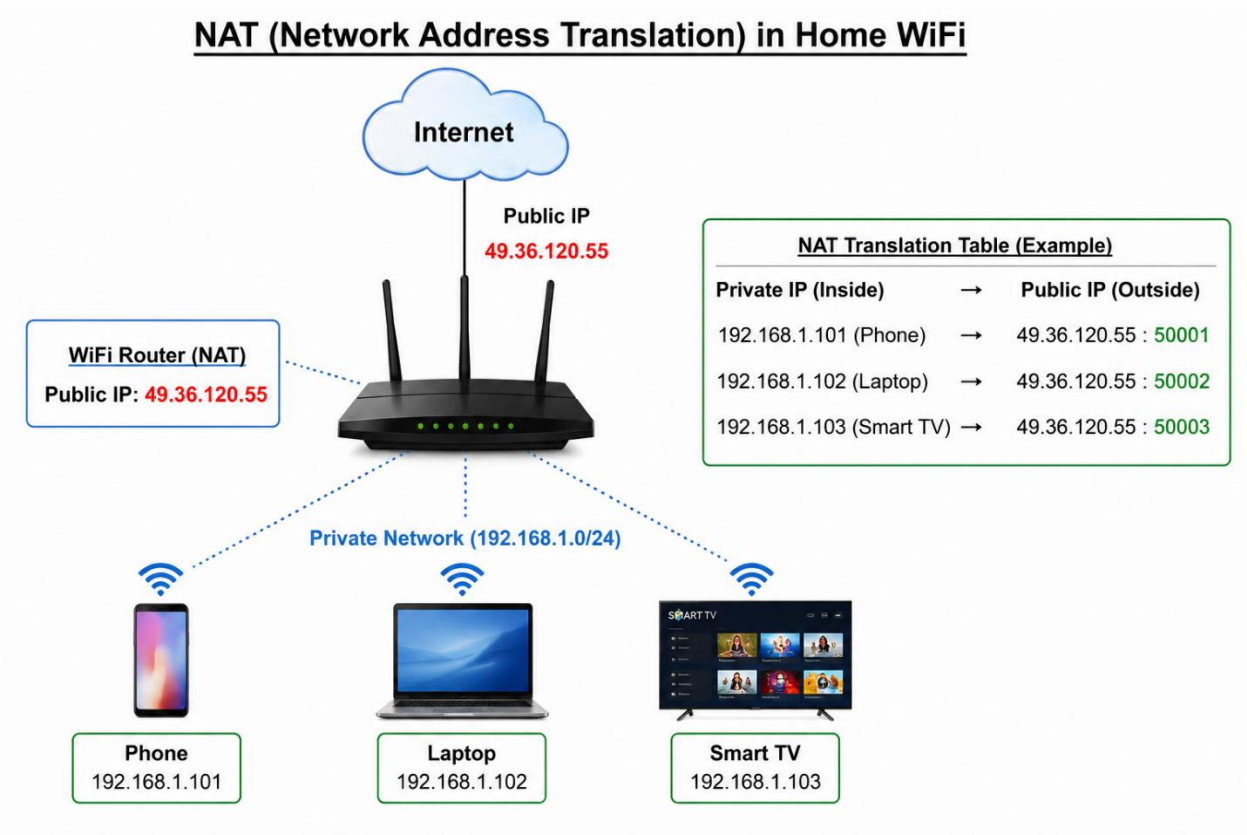
1.Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

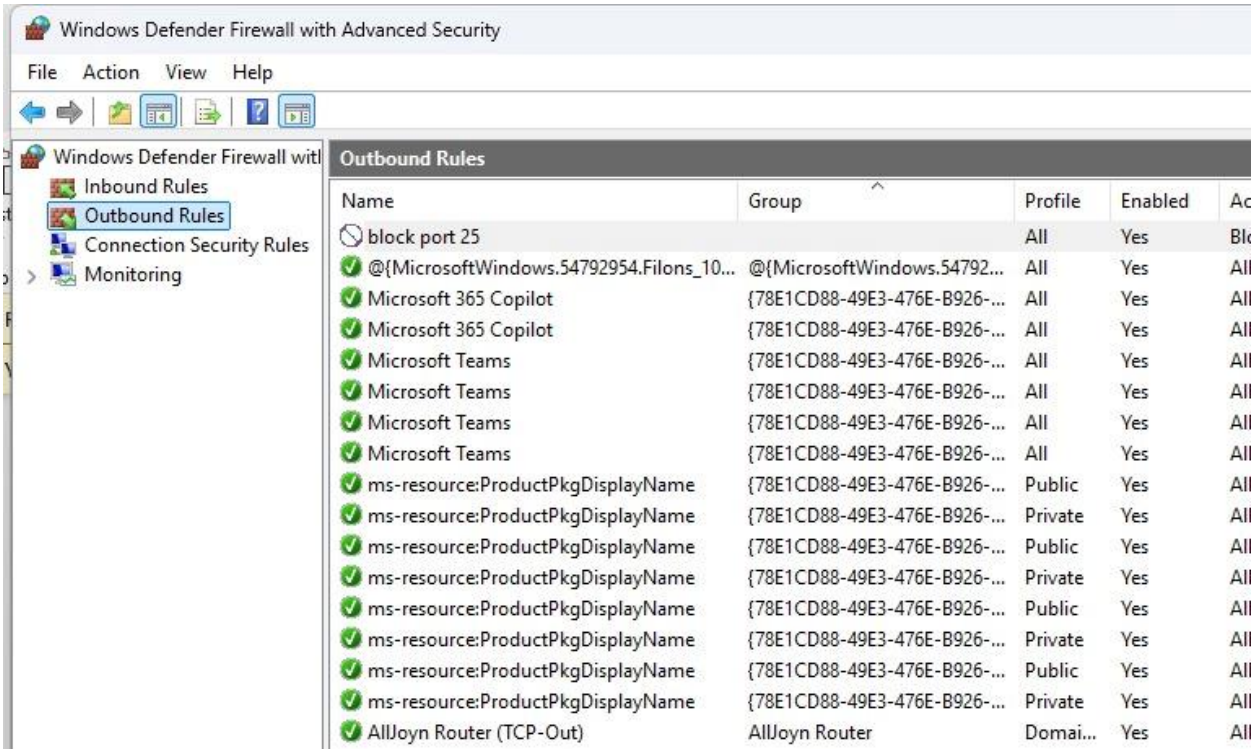


**2.Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.**



3.Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.



**4.Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.**

No	Packet	Direction/Port	Result	Reason
1	HTTP: 192.168.1.5 TO 172.217.166.78:80	HTTP/Port 80	Blocked	Port 80 is not explicitly allowed as default rule is block.
2	SSH request from 10.0.0.10 to 34.201.23.1:22	Outbound/Port22	Allowed	Rule explicitly says SSH outbound allow.
3	DNS query from 192.168.1.7 to 8.8.8.8:53	Outbound/Port53	Blocked	The rule only allows inbound DNS if it's secure. Outbound DNS is not listed so it remains default block.

5.Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Block unauthorized access – Firewalls block hackers from accessing Instagram or Zomato servers.

Stop DDoS attacks – Packet filtering blocks fake or suspicious traffic to keep the website or app running smoothly.

Allow only safe traffic – Firewalls allow trusted users and required ports (like HTTP/HTTPS) while blocking unwanted or harmful traffic.